

MEHADI HASAN

AI ENGINEER · FORWARD DEPLOYED

Senior Tech Lead · Solutions Architect

CONTACT

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AI / LLM STACK

- LLM & Agentic AI Systems
- RAG / Vector Search
- LangChain / LangGraph
- AWS Bedrock / Azure OpenAI
- n8n / Workflow Automation
- Prompt Engineering / Evals
- Fine-Tuning (LoRA / QLoRA)

TECH SKILLS

- HTML / CSS / JavaScript
- TypeScript / React / Vue
- Python / C / C++
- Node.js / NestJS
- Java / Kotlin / Go
- Flutter / Dart
- ASP.NET / C# / WPF
- PHP / Laravel
- REST / gRPC / GraphQL
- Kafka / RabbitMQ / MQTT

SOFT SKILLS

- Leadership
- Team Management
- Problem Solving

LANGUAGES

- English
- Bangla
- Hindi
- Malay (Basic)

15+ Years · Aviation Security · Government · Fintech · Healthtech

AI Engineering · Forward Deployed Engineering

PROFESSIONAL SUMMARY

Senior Tech Lead and **AI Engineer** with **15+ years** building mission-critical systems for **aviation security**, **government**, **fintech**, **healthtech** and **enterprise** clients. Works as a **Forward Deployed Engineer**: on site with the client, turning real operational problems into production systems. Currently **Sr. Tech Lead at Eleware Technology**, leading core modules of KLIA's airport security platforms (**PSIM, VHMS, AOP**), and in parallel leading 3 Malaysia Government transport programmes (LRT3, RTS, Smart Car Parking) at **Guildford Integrated System (GIS)**. Builds production **GenAI & Agentic AI**: **LLM integration** (OpenAI, Claude, Gemini, Ollama), **RAG**, **LangChain / LangGraph**, vector databases and **LLM fine-tuning with LoRA / QLoRA** (PEFT, bitsandbytes; own NeoHorse-Coder model) on **AWS Bedrock** and **Azure OpenAI**. Deep expertise in **system design (HLD/LLD)**, microservices, **event-driven architecture** (Kafka, RabbitMQ), **infrastructure as code with Ansible**, **n8n** automation and **MLOps / LLMOps**. Pursuing a **PhD in Computer Science**.

CORE COMPETENCIES

AI Engineering · Forward Deployed Eng. · LLM Integration · Agentic AI · LoRA / QLoRA · Fine-Tuning · RAG · Vector Search · LangChain · LangGraph · LlamaIndex · AWS Bedrock · Azure OpenAI · Vertex AI · MCP Servers · Prompt Engineering · LLM Evals · MLOps / LLMOps · n8n · HLD/LLD · Microservices · Event-Driven · Kafka · RabbitMQ · Real-time · Cloud Native · CI/CD · Ansible · Airport Security / PSIM · Fintech · Healthtech · Blockchain · Carbon Credit / ESG · OWASP Top 10 · AI Security · Open Source

PROFESSIONAL EXPERIENCE

Senior Tech Lead

Sep 2026 – Present

Eleware Technology Sdn. Bhd. – Airport Security Systems (KLIA) · Malaysia

- Lead core modules of 3 enterprise **airport security systems** at **KLIA**: **PSIM** (Physical Security Information Management), **VHMS** (Visitor & Vehicle Handling Management) and **AOP** (airport pass & permit platform)
- **PSIM:** one control-room console for alarms with SOP workflows, camera & door floor-plan maps, live/recorded video, cardholder tracking and server health – **.NET 9**, **ABP**, **Angular 20**, **SignalR** real-time push
- Integrate **Milestone XProtect** video, **Security Expert** access control, **under-vehicle scanners (UVSS)** and **SenseTime AI** face & appearance search; multi-node middleware with leased jobs for high availability
- **VHMS / AOP:** visitor & vehicle inspection, card printing, blacklist screening, e-vetting and AI face-quality checks on **.NET 10 / ABP**; mobile verifier app (.NET MAUI → Flutter)
- Production on **DC/DR** with **SQL Server Always On**; scripted deploy, rollback & health-check kits (PowerShell / SQL) for safe releases

Senior Software Engineer & Lead

Sep 2024 – Present

Guildford Integrated System (GIS) – Government Projects: LRT3, RTS, Smart Car Parking · Malaysia

- Lead engineer across **3 simultaneous Malaysia Government programmes**; own end-to-end HLD/LLD for **4-team, 18-engineer** organisation
- **LRT3:** C# .NET + PostgreSQL signalling backend processing **>2M events/day**; cut alarm noise ~**60%** via custom IP Scanner with multicast tracing
- **Smart Car Parking (TNG/Genting):** Flutter/Dart + NestJS + AI OCR; reduced gate latency from **4s** → **800ms** on integrated TNG/Mifare payment flow
- **MyDesk:** Custom AnyDesk-alternative remote desktop unifying control of all 3 programmes; eliminated 3 commercial-licence costs (~**RM 60k/yr saved**)
- Introduced **LLM-assisted log triage** (Claude/OpenAI) + **n8n** ops pipelines; cut mean-time-to-diagnose by ~**40%**
- Built **Kafka TUI** / **RabbitMQ TUI** (standard LRT3 ops diagnostics); SCADA via JMS, VMware ESXi, **Grafana** monitoring

Engineering Lead (Contract, Remote)

May – Oct 2025

Slotgator iGaming Project · Romania

- Delivered iGaming platform on **AWS + Kubernetes** sustaining ~**5k concurrent sessions**; zero P0 incidents over 6-month engagement
- Drove **Kafka** streaming + microservices, AES-256 payment hardening; led 7-engineer distributed team across 4 time zones

KEY INTERESTS

◇ Large Language Models
◇ Agentic AI & Autonomous Agents
◇ Generative AI Applications
◇ RAG & Vector Search
◇ LangChain / LangGraph
◇ AWS Bedrock / Azure OpenAI
◇ Model Context Protocol (MCP)
◇ n8n Workflow Automation
◇ Fintech, Healthtech & EdTech
◇ Blockchain & Web3
◇ Carbon Credit & ESG Tech
◇ MLOps & LLMOps
◇ Prompt Engineering & Evals
◇ Microservices Architecture
◇ Real-time Systems
◇ Cloud Native Development
◇ Cyber Security & R&D
◇ IoT & Embedded Systems

EXPERTISE

✓ AI Engineering
✓ Forward Deployed Engineering
✓ Airport Security (PSIM)
✓ AI / LLM System Design
✓ Agentic AI Workflows
✓ RAG Pipeline Engineering
✓ Full Stack Architecture
✓ System Design (HLD/LLD)
✓ Microservices
✓ Event-Driven Architecture
✓ API Development
✓ Cloud Infrastructure
✓ Real-time Systems
✓ CI/CD Pipelines
✓ Agile / Scrum Leadership
✓ Technical Consulting
✓ Team Mentorship
✓ Performance Optimization

COMPETENCIES

• LLM Integration
• Multi-Agent Orchestration
• Retrieval-Augmented Generation
• AI Pair Programming
• Machine Learning
• Distributed Systems
• Neural Networks

PLATFORMS

Unix, Linux, Windows, Android, iOS, macOS, AWS (Bedrock, EC2, S3, Lambda), Azure (OpenAI), GCP (Vertex AI), Docker, Kubernetes, Ansible, VMware ESXi

PROFESSIONAL EXPERIENCE (CONT.)

Full Stack Developer & DevOps Engineer *Dec 2023 – Sep 2024*
Gaming Industry · Malaysia & USA

- Shipped 4 hybrid iOS/Android apps reaching **>100k installs**; managed **AWS EC2**, Tencent & Alibaba Cloud infra
- Integrated 3 payment rails (Decoin, Alipay, UnionPay) with AES-256 + real-time WebSocket; CI/CD cut release cycle from 2w → 2 days

Project Manager / Team Lead *Dec 2022 – Oct 2023*
Creative IT Institute · Chittagong

- Led **12-engineer** cross-functional team across 5 client products using Agile/Scrum; delivered **100%** on-schedule release rate
- Designed PostgreSQL/MySQL schemas powering ~**30k DAU**; standardised Git, VPS & Nginx operating procedures

Team Development Lead & Sr. Full Stack *Mar – Oct 2022*
Datahostbd · Dhaka

- Mentored **8 junior engineers**; authored microservices training manual still in use today; architected microservices with API gateway + service discovery (SOLID); reduced cross-service latency ~**35%** vs. monolith baseline

Senior Consultant & Full Stack Developer *Mar 2020 – Jan 2022*
Creative IT Institute · Chittagong

- Hardened 6 production codebases through code review + unit-test rollout; reduced post-release defects ~**50%** YoY; coordinated deployments serving ~**8k students**; co-designed REST APIs with backend engineers

Chief Consultant & Engineer *Jan 2017 – Aug 2019*
TRZ · Gazipur

- Built end-to-end ERP for garments industry from scratch; RFID-integrated tracking across **4 factory floors**, **600+ daily operators**
- Automated scheduled power management saving ~**18%** electricity cost; BI model in Matplotlib/Pandas/NumPy adopted by exec leadership

CEO & Chief Consultant *Jul 2011 – Nov 2016*
Orimpo Inc. Ltd · Dhaka

- Founded & ran consultancy for 5+ years; delivered **40+ enterprise projects** on J2EE/JSP/To across govt, fintech, retail; ran security assessments & penetration testing
- Led **bKash mobile payment prototyping** with Robi Axiata (Java) – foundational work for Bangladesh's largest mobile-money platform

EDUCATION

Doctor of Philosophy (PhD) – Computer Science *Jul 2025 – Ongoing*
Lincoln University College · Part-time

Master of Computer Science *Sep 2023 – May 2025*
Lincoln University College

Thesis: AI-Driven Medical Assistant

BSc in Computer Science & Engineering *2009 – 2017*
United International University

Project: *An Intelligent Building Power Management System (IoT)*

ADDITIONAL CONSULTANCY

- **Alamin Sweaters** *(Feb – May 2013)* – Garments production workflow optimisation & barcode tracking
- **Friends Appeals** *(Mar – Aug 2018)* – Business management software, DB architecture & reporting
- **Peregrine Air Line (UK)** *(Jan – Jun 2018)* – E-commerce, stock & shipment tracking, payment gateway

METHODOLOGY

- Scrum / Agile / Waterfall
- SOLID Principles
- Domain-Driven Design
- Redmine / Jira
- Git / Bitbucket
- Jenkins CI/CD
- Test-Driven Development
- LLM Evals & Red-Teaming

DATABASES

Relational: Oracle, MSSQL, MySQL, PostgreSQL, SQLite

NoSQL: MongoDB, Redis, Firebase

Vector: Pinecone, Weaviate, Qdrant, Chroma, FAISS, Milvus, pgvector

AI / GENAI STACK

LLM APIs: OpenAI, Claude, Gemini, Grok, Mistral

Self-Hosted: Ollama, vLLM, llama.cpp

Orchestration: LangChain, LangGraph, LlamaIndex

Agents: MCP, Function Calling, ReAct, AutoGen

Cloud LLM: AWS Bedrock, Azure OpenAI, Vertex AI

Fine-tuning: LoRA, QLoRA, PEFT, MLX, GGUF

Eval: LangSmith, Langfuse, Promptfoo, RAGAS

DEV TOOLS

VS Code, **Neovim**, JetBrains, OWASP ZAP, **n8n**, **Grafana**, **RabbitMQ**, Ollama, **Claude Code**, **GitHub Copilot**, **Gemini Code Assist**, **OpenAI Codex**, **LaTeX**, **Overleaf**

Custom Tools: **MetaSig**, **KafkaTUI**, **RabbitMQTUI**, **IPScanner** (custom network mapping & visualization), custom ticket monitoring

Networking: Tailscale VPN, OpenVPN, WireGuard, multicast tracing

PORTFOLIO VIDEOS

- ▷ LRT / MRT / RTS Signaling
- ▷ Project Management
- ▷ Inventory System
- ▷ UK Export / Import
- ▷ Smart Building IoT
- ▷ Gaming Project
- ▷ iGaming Project

SELECTED PROJECTS

▶ **MetaSig – Cross-Platform Remote Device Management Platform** *(2025 – Present)* – Production fleet-management TUI for Android/iOS/macOS/Win/Linux: WebSocket signalling, remote shell, file browser, live screen preview, APK packaging & signing pipeline. Manages **>200 endpoints** across 3 Malaysia Gov programmes from a single terminal. *Stack:* Node.js, neo-blessed TUI, WebSocket, Android agent

▶ **NeoHorse-Coder-4B – Custom Fine-Tuned Coding LLM (QLoRA)** *(Sep 2026 – Present)* – Own offline patch / fix / code-review model for local AI agents (OpenCode + custom model router), built over **35+ training iterations** on a 4B hybrid linear-attention base (Qwen3.5 family). **QLoRA** 4-bit NF4 (PEFT + bitsandbytes, r=32, α=64) on **5.4k** curated examples from own commits, CommitPackFT & agent traces; trained on RunPod A40 & AWS L4 GPUs; exported to GGUF (llama.cpp, Q4_K_M). Held-out eval vs base: grounded agent edits **10%** → **91%**, right-tool use **34%** → **94%**; speculative decoding **15** → **45 tok/s**; v34 test-writing specialist. *Stack:* PyTorch, Transformers, PEFT, MLX, llama.cpp

▶ **GPT-5.3 Agent Bridge + Pentest-AI Red-Team Deployment** *(2025 – Present)* – Built OpenAI-compatible HTTP bridge wrapping ChatGPT GPT-5.3 via Playwright; integrated as multi-model agent inside **opencode** TUI for agentic dev workflows. *Separately:* authorised internal red-team deployment of upstream open-source Pentest-AI (ptai, original by 0xSteph) for OWASP Top 10 exercises via MCP – *integration role, not authorship*

▶ **Smart Car Parking System (TNG / Genting)** *(Sep 2024 – Present)* – AI OCR plate-detection + C/C++ hardware gate control + TNG/Mifare payment integration; cut gate-pass latency from **4s** → **800ms**. *Stack:* Flutter/Dart, NestJS, C# WinForms (CPMS SAS Scheduler), Python OCR, PostgreSQL

▶ **LRT3 / RTS Signalling Systems + KafkaTUI / RabbitMQ-TUI** *(Sep 2024 – Present)* – Mission-critical Malaysia Gov. transport signalling processing **>2M events/day** (C# .NET + PostgreSQL + Java/PHP). Open-source TUI diagnostics now standard for LRT3 ops team; cut alarm noise ~**60%**

▶ **Medical AI Assistant (MSc Thesis) + AI Doctor Triage Platform** *(2024 – 2025)* – HIPAA-aware healthcare RAG over clinical knowledge base (Claude / OpenAI / Ollama, LangChain, LangGraph, Pinecone/Qdrant/pgvector, FastAPI). AI Doctor: Gemini-powered triage with adaptive symptom intake; React Native + Node.js + Prisma; Dockerised

▶ **Open-Source Suite – 16+ public repos** *(2023 – 2026)* – mydesk (AnyDesk-alternative), macscp (Swift SFTP), snmp-alarm (NOC alarming), SignalR→WS bridge, nginx-RTMP for macOS, SuperCharge SG (Telegram RAG bot), Free LLM API Keys, 6 Redmine plugins (forum-pdf, doc-pdf, turbo-nav, issue-emailer, dark-mode, custom-mail)

▶ **Earlier flagship work** *(2015 – 2024)* – P2P Remote Desktop (C++/Qt + WebRTC, H.264/VP8); Custom Dev TUI Suite (multi-framework scaffolder, audio editor, DB exporter, .deb packager); Job-Search AI (9-source aggregator, AI cover letters, SMTP apply); n8n / Ollama infra; Telegram Dating Bot (Stripe + Drizzle); Hathe Khori (TF/Flutter); DMP crime clustering; RFID ERP + encrypted IRC for Bangladesh Army

CERTIFICATIONS & ACHIEVEMENTS



Master of Computer Science
Lincoln Univ. – 2025



Microsoft Imagine Cup
2012



Programming Contest
Qazi Azher Ali – 2011



UIU Robotics Workshop
2011



NYDSA Certificate
National Youth Dev.



Code Warriors
Programming Comp.

SPEAKING, MENTORSHIP & COMMUNITY

■ Mentored junior developers across Bangladesh & Malaysia teams; wrote training manuals on microservices, SOLID & secure coding

■ Active contributor to **Generative AI / Agentic AI** community via public repos (KafkaTUI, RabbitMQ-TUI, Free LLM API Keys)

■ Internal knowledge-sharing sessions on **AI-assisted software engineering** & mission-critical signalling architecture for Malaysia Government programme teams

Thank you for reviewing my profile.

Looking forward to connecting with you.